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Integral Approach of Unsupervised Learning and Temporal Stability in Irrigation Management

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ABSTRACT. *A detailed understanding of variation in soil moisture is needed for developing precision irrigation strategies that can maximize crop production while minimizing consumptive water use. The overarching aim of the study was to optimize the representative location for precision water management with the integration of unsupervised learning and temporal stability analyses. The hypothesis for this objective was that a zone-specific representative sensor location can be determined for variable rate irrigation management in the field. The study was conducted in Tennessee Valley Region of Northern Alabama, USA. The study found that each zone in the field can be represented with a temporally stable location to determine the average zone-specific soil moisture using the concept of unsupervised learning for management zone delineation and temporal stability in soil moisture, which showed that the zone-specific irrigation can be scheduled during the growing season. Farmers can use the dry, wet, and temporally stable locations for establishing the best irrigation scheduling to increase water-use efficiency during the growing season.*

Keywords. *Unsupervised learning, irrigation management zones, soil water retention curve, HYPROP, variable rate irrigation*

INTRODUCTION

Irrigation is an influential aspect of the water cycle in terms of fresh water-use. Nearly 20% of the irrigated area of the world's total cultivated area meets 40% of the global food supply (Filippucci et al., 2020). The global food demand should be expected to increase significantly due to the rapid increase in

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the population from approximately 7.9 billion in 2022 to nearly 9 billion by 2050 with an average annual growth rate of 0.9% (U.S. Census Bureau, 2022). Due to climate change and anticipated variability in the distribution and timing of rainfall, much agriculture is shifting towards irrigated agriculture, including in humid regions (Kumar et al., 2022a). Recently, for example, Takhellambam et al. (2022a, 2022b) forecasted shifts in the rainfall pattern at the sub-daily interval until 2059 in the Southeastern USA, which was shown to not only affect the frequency of rainfall events such as extreme rainfall, extreme droughts, rainfall erosivity (Takhellambam et al., 2022c, 2023), but also climate- and water-smart irrigation scheduling (Kumar et al., 2023). Therefore, we need more climate-resilient solutions around site-specific irrigation for precision water management in the face of changing climate. Different irrigation techniques, such as sensor-based irrigation and variable rate irrigation (VRI) can be considered best irrigation management strategies (Kumar et al., 2022a, 2023) to meet growing freshwater requirements, to avoid over and under application of water, to increase water-use efficiency, and to meet the crop water demands based on need (Filippucci et al., 2020).

To adopt VRI as a best irrigation management practice in cropped fields, a field can be delineated in different management zones (MZs) based on crop yield variability, soil type, and topography (Kumar et al., 2021a, 2022a, 2022b, 2023). Delineating MZs has the potential to help conserve water by more closely meeting crop water demands based on zones (Liakos et al., 2017). However, precise irrigation to meet zone-specific crop water demand requires continuous monitoring of soil water dynamics within the zones (Fontanet et al., 2020). Further, real-time soil moisture information (Reyes et al., 2019) would be needed at different soil depths due to uneven dryness and wetness in order to manage zone-specific irrigation efficiently at the right time and the right place during the growing season. Although previous studies have indicated the need for research defining irrigation management zones (Filho et al., 2020; Kumar et al., 2023) and exploring soil moisture temporal stability (Kumar et al., 2021b), no study to date has investigated the temporally stability concept in conjunction with irrigation management zones. In this study, we are combining both concepts to come up with innovative and efficient irrigation scheme to adopt in the crop fields. As such, the goal of this study is to develop a method for best irrigation management practice by integrating management zone analysis and temporal stability concept within the crop field.

MATERIALS AND METHODS

Study site description and management zones delineation

The study was conducted in a 120 ha corn field located in Town Creek Watershed in the Tennessee Valley region of North Alabama, USA (Fig. 1). The study area is in the subtropical humid climate

region with an average annual rainfall of 1400 mm, including an average rainfall of 570 mm during the growing season (1951-2017). To accomplish the goal of this study, the field was set up with Reinke center pivot irrigation system (Reinke Manufacturing Co., Inc., Deshler, NE, USA), which was retrofitted with variable rate irrigation (VRI) technology. A catch can test was conducted to evaluate and ensure the uniformity application of the center pivot irrigation system (Ortiz et al., 2021). The field was delineated in three different irrigation management zones (Fig. 1) using Management Zone Analyst (MZA, USDA-ARS, Columbia, MO, USA) software, which uses fuzzy c-means unsupervised clustering algorithm and Mahalanobis distance that allocates field information into homogeneous areas or management zones (Fridgen et al., 2004). Multiple Watermark soil moisture sensors were installed at 15 cm, 30 cm, and 60 cm soil depths in each zone. The sensors recorded soil matric potentials, which was converted into soil moisture content using the van Genuchten model (van Genuchten, 1980).

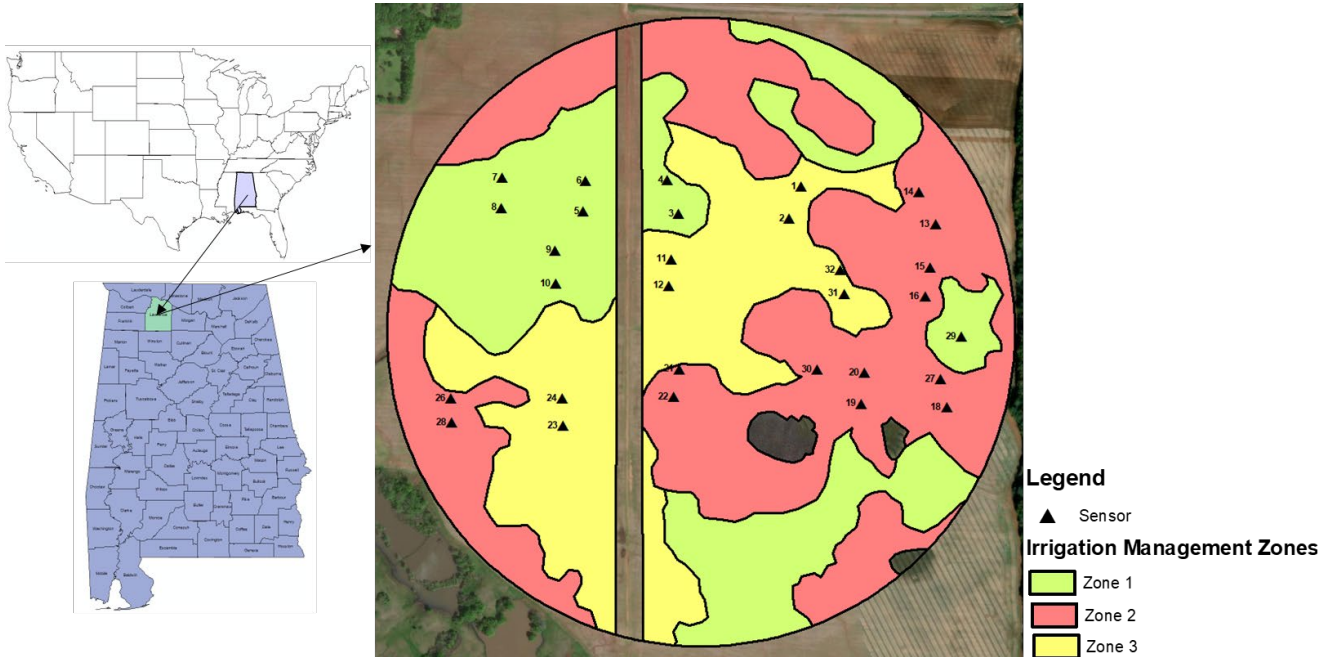


Fig. 1. Schematic diagram of the study field with irrigation management zones and sensor locations

Temporal stability analysis

The temporal stability in soil moisture determines the representative location in the field based on in-situ measurements (Vachaud et al., 1985). We used different statistical approaches to estimate the temporal stability in each zone. The mean relative difference $\bar{\delta}_i$ was defined as

$$\bar{\delta}_i = \frac{\sum_{j=1}^N \bar{\delta}_{ij}}{N} \quad (1)$$

where $\bar{\delta}_{ij}$ is relative difference in soil moisture at each location (i) and each time (j), N number of days soil moisture data was collected. The mean relative difference quantifies the departure in soil moisture

of a location from the average zone-specific soil moisture, which indicates if any location is drier or wetter on a day compared to the average zone-specific soil moisture in the respective zone.

The standard deviation of relative difference $\sigma(\delta_i)$ was defined as

$$\sigma(\delta_i) = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (\delta_{ij} - \bar{\delta}_i)^2} \quad (2)$$

A temporally stable or representative location has a small $\sigma(\delta_i)$ and a mean relative difference close to zero. A location with a greater or smaller value than zero for mean relative difference indicates wetter or drier conditions, respectively.

A single statistic, the index of time stability (ITS), was proposed by Jacobs et al. (2004) to determine temporal stability, which represents the best sensor location for measuring in-situ soil moisture as the average zone-specific soil moisture. A smaller value of ITS for a location indicates the temporally stable location in the zone. The ITS can be defined per location as

$$ITS_i = \sqrt{(\bar{\delta}_i)^2 + \sigma(\delta_i)^2} \quad (3)$$

Sometimes, these statistics can mislead the outcomes of the temporal stability to define a representative location. Therefore, the temporal stability in soil moisture was validated using the linear regression relationship between spatial mean of zone-specific soil moisture and soil moisture of temporally stable location in the zone.

RESULTS AND DISCUSSION

Zone-specific variability

In zone 1, the $|\bar{\delta}_i|$ varied from 0.002 to 0.041 at 15 cm soil depth, from 0.003 to 0.067 at 30 cm soil depth, and from 0.013 to 0.123 at 60 cm soil depth. The $\sigma(\delta_i)$ values ranged from 0.042 to 0.074 at 15 cm soil depth, from 0.038 to 0.069 at 30 cm soil depth, and from 0.042 to 0.106 at 60 cm soil depth. The ITS varied from 0.045 to 0.083 at 15 cm soil depth, from 0.040 to 0.092 at 30 cm soil depth, and from 0.047 to 0.162 at 60 cm soil depth.

In zone 2, the $|\bar{\delta}_i|$ varied from 0.001 to 0.124 at 15 cm soil depth, from 0.002 to 0.055 at 30 cm soil depth, and from 0.022 to 0.127 at 60 cm soil depth. The $\sigma(\delta_i)$ values ranged from 0.035 to 0.108 at 15 cm soil depth, from 0.018 to 0.040 at 30 cm soil depth, and from 0.024 to 0.048 at 60 cm soil depth.

The ITS varied from 0.046 to 0.136 at 15 cm soil depth, from 0.020 to 0.068 at 30 cm soil depth, and from 0.053 to 0.133 at 60 cm soil depth.

In zone 3, the $|\bar{\delta}_i|$ varied from 0.001 to 0.070 at 15 cm soil depth, from 0.026 to 0.069 at 30 cm soil depth, and from 0.008 to 0.138 at 60 cm soil depth. The $\sigma(\delta_i)$ values ranged from 0.038 to 0.061 at 15 cm soil depth, from 0.032 to 0.052 at 30 cm soil depth, and from 0.062 to 0.091 at 60 cm soil depth. The ITS varied from 0.050 to 0.093 at 15 cm soil depth, from 0.045 to 0.084 at 30 cm soil depth, and from 0.067 to 0.156 at 60 cm soil depth.

The variability in the temporal stability statistics existed due to topographic variability in the irrigation management zones and zone-specific properties responsible for water movement and soil water storage in the soil profiles (Kumar et al., 2022a, 2023).

Zone-specific temporal stability

Temporal stability statistics were used to find the representative sensor location to estimate the spatial mean in zone-specific soil moisture. In zone 1, soil moisture at sensor S9 location was found to be temporally stable at 15 cm, 30 cm, and 60 cm soil depths (Figs. 2a-2c). The S9 location can thus be considered as a representative location to determine the average zone-specific soil moisture in zone 1. At all depths, we also developed the linear regression relationship between soil moisture at S9 and spatial mean of soil moisture in zone 1 (Figs. 2d-2f). The linear regression proved that S9 location represents the average soil moisture in zone 1 for a variably irrigated irrigation field. The S9 location had soil properties and topographic gradients that were identical to the average soil properties in zone 1. Similar outcomes were also observed by Kumar et al. (2021b) in a uniformly irrigated field having uniform soil and topographic characteristics of least slope in the field. Based on the temporal stability statistics, we found that the locations S6 and S10 were always dry, while the locations S3, S4, S5, S7, and S8 were always wet at 15 cm soil depth when compared to the S9 location in zone 1. At 30 cm soil depth, the S6, S7, and S8 locations were always dry, while the S3, S4, S5, and S10 locations were always wet when compared to the S9 location in zone 1. When comparing sensors with the S9 location at 60 cm soil depth in zone 1, S3, S4, S6, S7, S8, and S10 locations were always dry, while the S5 location was always wet in zone 1.

It should be noted, however, that none of the sensors were far from the temporally stable sensor S9 with regards to their overall dryness and wetness condition. This variation in dryness and wetness showed soil heterogeneity within zone 1 from location to location, which was captured by a temporally stable location. Sensor S9 was capable of addressing most of the soil moisture variability within the zone that could be affected due to small changes in the zone-specific topographic attributes and soil properties.

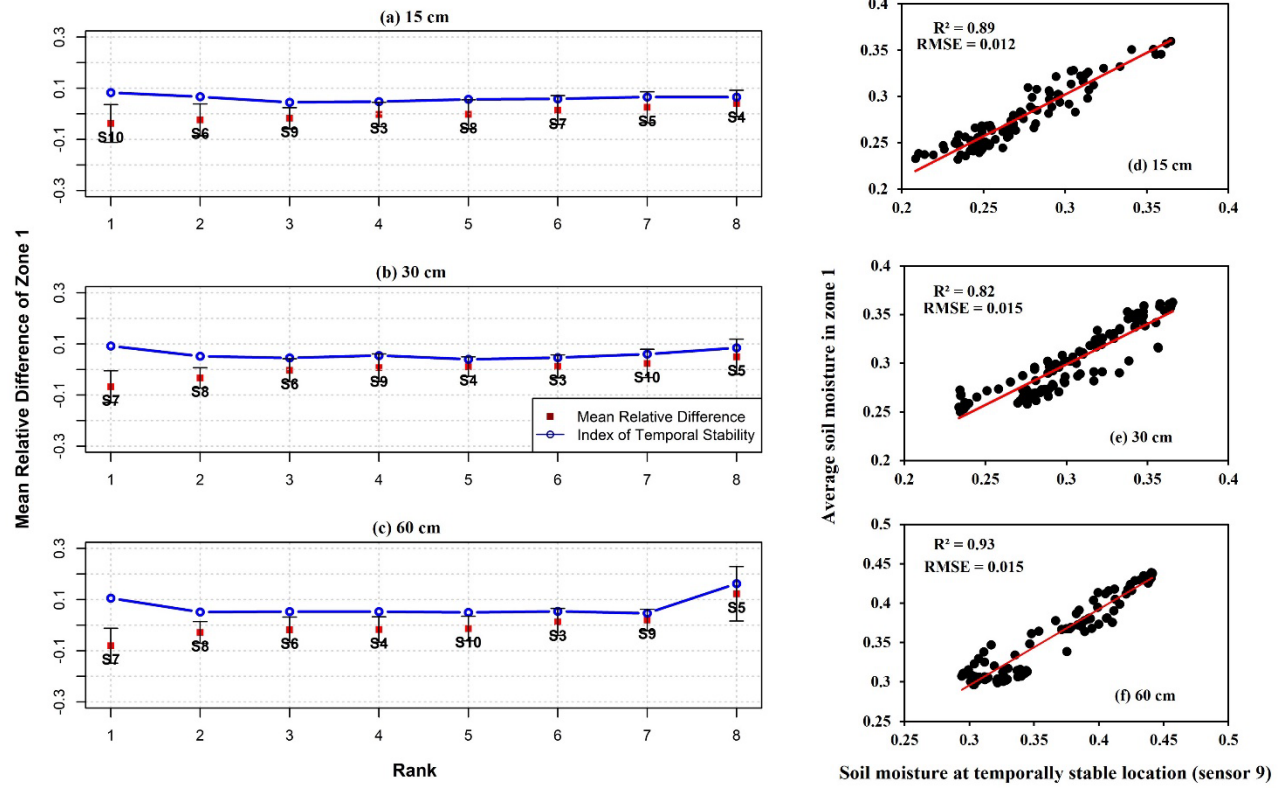


Figure 2. Plots of rank ordered mean relative difference and index of temporal stability of soil moisture and error bars show standard deviations in relative difference in zone 1(a-c). Plots show linear regression relationship between average soil moisture and temporally stable location in zone 1 at different soil depths (d-f).

In zone 2, soil moisture at sensor S20 location was found to be temporally stable at 15 cm, 30 cm, and 60 cm soil depths (Figs. 3a-3c). The S20 location was under-representing the average zone-specific soil moisture at 60 cm soil depth, however, soil moisture at S20 location still indicated better than other sensor locations in zone 2. The topographic variability in the zone caused the lateral flow as surface and subsurface runoff, which resulted in reduced vertical infiltration and lower soil moisture at S20 location than average zone-specific soil moisture. The S20 location can thus be considered as a representative location to determine the average zone-specific soil moisture in zone 2. At all depths, we also developed the linear regression relationship between soil moisture at S20 location and spatial mean of soil moisture in zone 2 (Figs. 3d-3f). The linear regression proved that S20 location represents the average soil moisture in zone 2 for a variably irrigated irrigation field. The S20 location had similar soil properties and topographic gradients that were identical to the average soil properties in zone 2. Similar outcomes were also observed by Kumar et al. (2021b) in a uniformly irrigated field with higher slope and topographic characteristics. Based on the temporal stability statistics, we found that the locations S13, S14, S15, S16, S19, S21, and S10 were always dry, while the locations S18, S22, S27, and S30 were

always wet at 15 cm soil depth when compared to the S20 location in zone 2. At 30 cm and 60 cm soil depths, the S19 location was always dry, while the S13, S14, S15, S16, S18, S21, S22, S27, and S30 locations were always wet when compared to the S20 location in zone 2. However, none of the locations were far from the temporally stable S20 location in the condition of dryness and wetness.

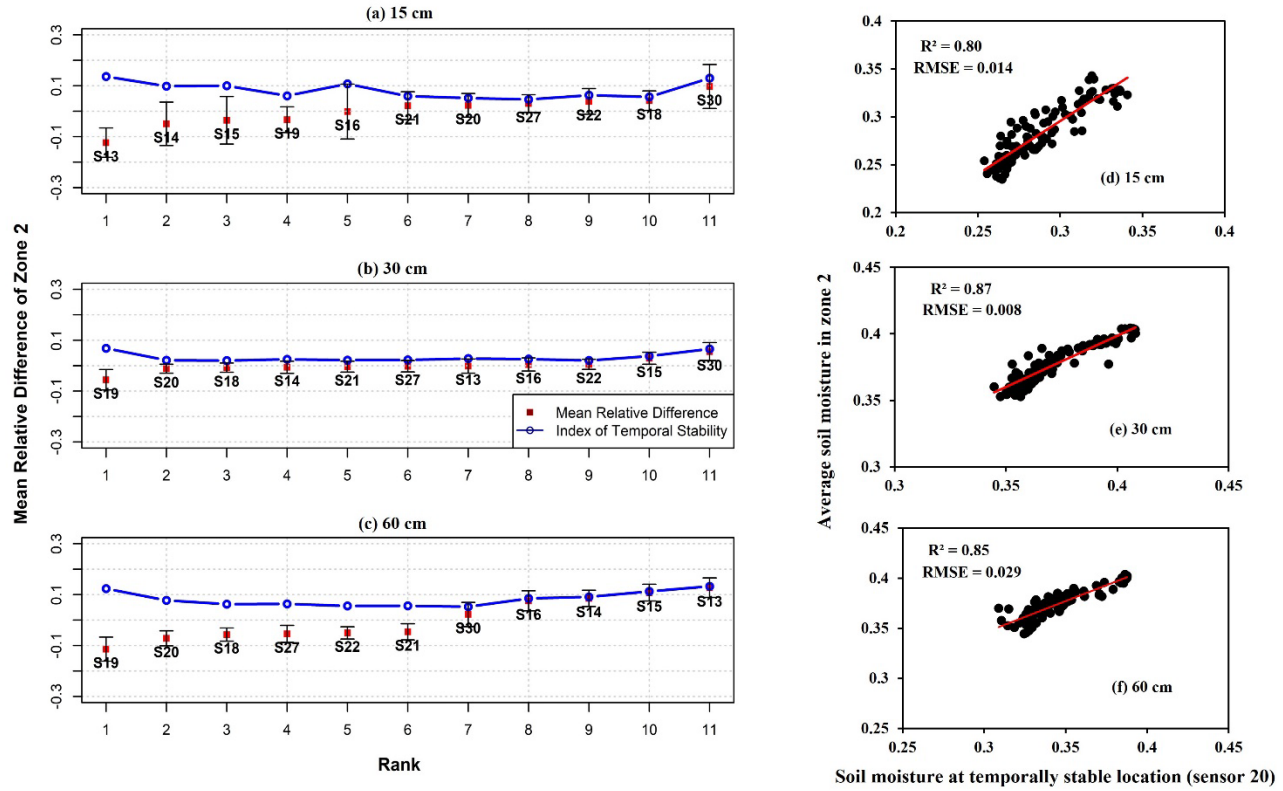


Figure 3. Plots of rank ordered mean relative difference and index of temporal stability of soil moisture and error bars show standard deviations in relative difference in zone 2 (a-c). Plots show linear regression relationship between average soil moisture and temporally stable location in zone 1 at different soil depths (d-f).

In zone 3, soil moisture at sensor S23 location was found to be temporally stable at 15 cm, 30 cm, and 60 cm soil depths (Figs. 4a-4c). The S23 location can thus be considered as a representative location to determine the average zone-specific soil moisture in zone 3. The linear regression relationship between soil moisture at S23 location and spatial mean of soil moisture in zone 3 showed a strong R^2 and lower RMSE at 15 cm, 30 cm, and 60 cm soil depths (Figs. 4d-4f) when compared to soil moisture at other locations. The linear regression proved that S23 location represents the average soil moisture in zone 1 for a variably irrigated irrigation field. The S23 location had soil properties and topographic gradients that were identical to the average soil properties in zone 3. Based on the temporal stability statistics, we found that the S12 and S32 locations were always dry, while the locations S1, S2, S11, S24, and S31 were always wet at 15 cm soil depth when compared to S23 location in zone 3. At 30 cm soil depth,

S32 location was always dry, while the locations S1, S2, S11, S12, S24, and S31 were always wet when compared to S23 location in zone 3. When comparing different locations with S23 location at 60 cm soil depth in zone 3, the locations S11, S12, and S32 were always dry, while the locations S1, S2, S24, and S31 were always wet in zone 3. However, none of the locations were far from the temporally stable location S23 in the condition of dryness and wetness. This variation in dryness and wetness showed soil heterogeneity within zone 3 from location to location, which was captured by a temporally stable location. The location S23 was capable of addressing most of the soil moisture variability within the zone that could be affected due to small changes in the zone-specific topographic attributes and soil properties.

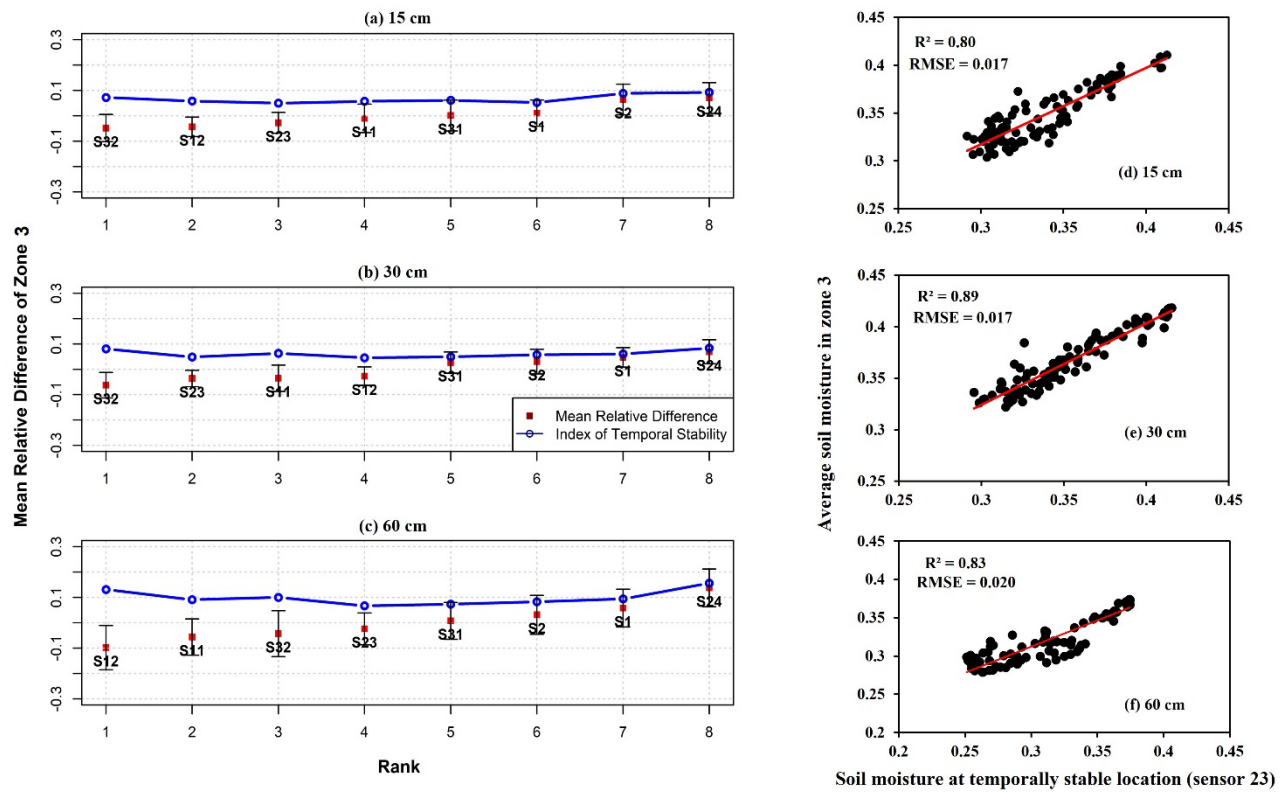


Figure 4. Plots of rank ordered mean relative difference and index of temporal stability of soil moisture and error bars show standard deviations in relative difference in zone 3 (a-c). Plots show linear regression relationship between average soil moisture and temporally stable location in zone 1 at different soil depths (d-f).

CONCLUSION

The study was conducted with the integration of management zone delineation and temporal stability concepts. The field was delineated in three different irrigation management zones based on the soil texture, topography, and historical crop yield. In each zone, a temporally stable location was identified,

which can be used to predict the average zone-specific soil moisture at soil depths. Knowing the most temporal stable location in each zone, an accurate determination of time and rate of irrigation can lead to precision variable rate irrigation, ultimately resulting in effective irrigation water management. Since the field varies in soil, topography, and crop yield, management zones can be delineated, and for adopting a variable rate irrigation scheme, temporal stability can help determine the average soil moisture of the zone to trigger the irrigation. For sustainable water resources and irrigation management in a changing climate, zone-specific optimized solutions can be potential approaches in the future.

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REFERENCES:

- Filippucci, P., Tarpanelli, A., Massari, C., Serafini, A., Strati, V., Alberi, M., Raptis, K. G. C., Mantovani, F., & Brocca, L. (2020). Soil moisture as a potential variable for tracking and quantifying irrigation: A case study with proximal gamma-ray spectroscopy data. *Advances in Water Resources*, 136, 103502. <https://doi.org/10.1016/j.advwatres.2019.103502>
- Filho, J. F. D. C. L., Ortiz, B. V., Damianidis, D., Balkcom, K. S., Dougherty, M., & Knappenberger, T. (2020). Irrigation scheduling to promote corn productivity in central Alabama. *J. Agric. Sci.*, 12(9). <https://doi.org/10.5539/jas.v12n9p34>
- Fontanet, M., Scudiero, E., Skaggs, T. H., Fernández-Garcia, D., Ferrer, F., Rodrigo, G., & Bellvert, J. (2020). Dynamic Management Zones for Irrigation Scheduling. *Agricultural Water Management*, 238, 106207. <https://doi.org/10.1016/j.agwat.2020.106207>
- Fridgen, J. J., Kitchen, N. R., Sudduth, K. A., Drummond, S. T., Wiebold, W. J., & Fraisse, C. W. (2004). Management Zone Analyst (MZA). *Agron. J.*, 96(1), 100-108. <https://doi.org/10.2134/agronj2004.1000>
- Jacobs, J. M., Mohanty, B. P., Hsu, E.-C., & Miller, D. (2004). SMEX02: Field-scale variability, time stability, and similarity of soil moisture. *Remote Sens. Environ.*, 92(4), 436-446. <https://doi.org/10.1016/j.rse.2004.02.017>
- Kumar, H., Srivastava, P., Lamba, J., Lena, B., Diamantopoulos, E., Ortiz, B., Takhellambam, B., Morata, G., & Bondesan, L. (2023). A methodology to optimize site-specific field capacity and irrigation thresholds. *Agricultural Water Management*, 286, 108385. <https://doi.org/10.1016/J.AGWAT.2023.108385>
- Kumar, H., Srivastava, P., Lamba, J., Diamantopoulos, E., Ortiz, B., Morata, G., Takhellambam, B., & Bondesan, L. (2022a). Site-specific irrigation scheduling using one-layer soil hydraulic properties and inverse modeling. *Agricultural Water Management*, 273, 107877. <https://doi.org/10.1016/J.AGWAT.2022.107877>
- Kumar, H., Srivastava, P., Lamba, J., Ortiz, B. V., Way, T. R., Sangha, L., Takhellambam, B. S., Morata,

- G., & Molinari, R. (2022b). Within-Field Variability in Nutrients for Site-Specific Agricultural Management in Irrigated Cornfield. *Journal of the ASABE*, 65(4), 865–880. <https://doi.org/10.13031/JA.15042>
- Kumar, H., Srivastava, P., Lamba, J., Ortiz, B. V., Way, T. R., Sangha, L., ... & Morata, G. (2021a). Phosphorus Variability in the Irrigated Cropland During a growing season. *Proc. 2021 ASABE Annual International Virtual Meeting*. ASABE. <https://doi.org/10.13031/AIM.202100886>
- Kumar, H., Srivastava, P., Ortiz, B. V., Morata, G., Takhellambam, B. S., Lamba, J., & Bondesan, L. (2021b). Field-Scale Spatial and Temporal Soil Water Variability in Irrigated Croplands. *Transactions of the ASABE*, 64(4), 1277–1294. <https://doi.org/10.13031/TRANS.14335>
- Liakos, V., Porter, W., Liang, X., Tucker, M. A., McLendon, A., & Vellidis, G. (2017). Dynamic Variable Rate Irrigation – A Tool for Greatly Improving Water Use Efficiency. *Advances in Animal Biosciences*, 8(2), 557–563. <https://doi.org/10.1017/S2040470017000711>
- Ortiz, B. V., Bondesan, L., Morata, G., McClendon, P., & Kumar, H. (2021). Maintaining water application uniformity in irrigation systems. Alabama Cooperative Extension System. Retrieved from <https://www.aces.edu/blog/topics/crop-production/maintaining-water-application-uniformity-in-irrigation-systems/>
- Reyes, J., Wendroth, O., Matocha, C., & Zhu, J. (2019). Delineating Site-Specific Management Zones and Evaluating Soil Water Temporal Dynamics in a Farmer's Field in Kentucky. *Vadose Zone Journal*, 18(1), 180143. <https://doi.org/10.2136/vzj2018.07.0143>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R. P., Kumar, H., & Tian, D. (2023). Projected mid-century rainfall erosivity under climate change over the southeastern United States. *Science of the Total Environment*, 865, 2–9. <https://doi.org/10.1016/j.scitotenv.2022.161119>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R. P., Kumar, H., & Tian, D. (2022a). Temporal disaggregation of hourly precipitation under changing climate over the Southeast United States. *Scientific Data* 2022 9:1, 9(1), 1–14. <https://doi.org/10.1038/s41597-022-01304-7>
- Takhellambam, B. S., Srivastava, P., Lamba, J., Zhao, W., Kumar, H., & Tian, D. (2022b). Assessment of projected change in Intensity-duration-frequency (IDF) curves for Southeastern, United States using Artificial Neural Networks. *Proc. 2022 ASABE Annual International Meeting*, ASABE. <https://doi.org/10.13031/AIM.202200175>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R. P., Kumar, H., & Tian, D. (2022c). Projected Rainfall Erosivity Under Climate Change in the Southeastern United States. *Proc. 2022 ASABE Annual International Meeting*, ASABE. <https://doi.org/10.13031/aim.202200176>
- U.S. Census Bureau (2022). Population and Housing Unit Estimates. Retrieved from <https://www.census.gov/programs-surveys/popest.html>
- Vachaud, G., Passerat De Silans, A., Balabanis, P., & Vauclin, M. (1985). Temporal stability of spatially measured soil water probability density function. *SSSA J.*, 49(4), 822-828. <https://doi.org/10.2136/sssaj1985.03615995004900040006x>
- van Genuchten, M. T. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *SSSA J.*, 44(5), 892-898. <https://doi.org/10.2136/sssaj1980.03615995004400050002x>